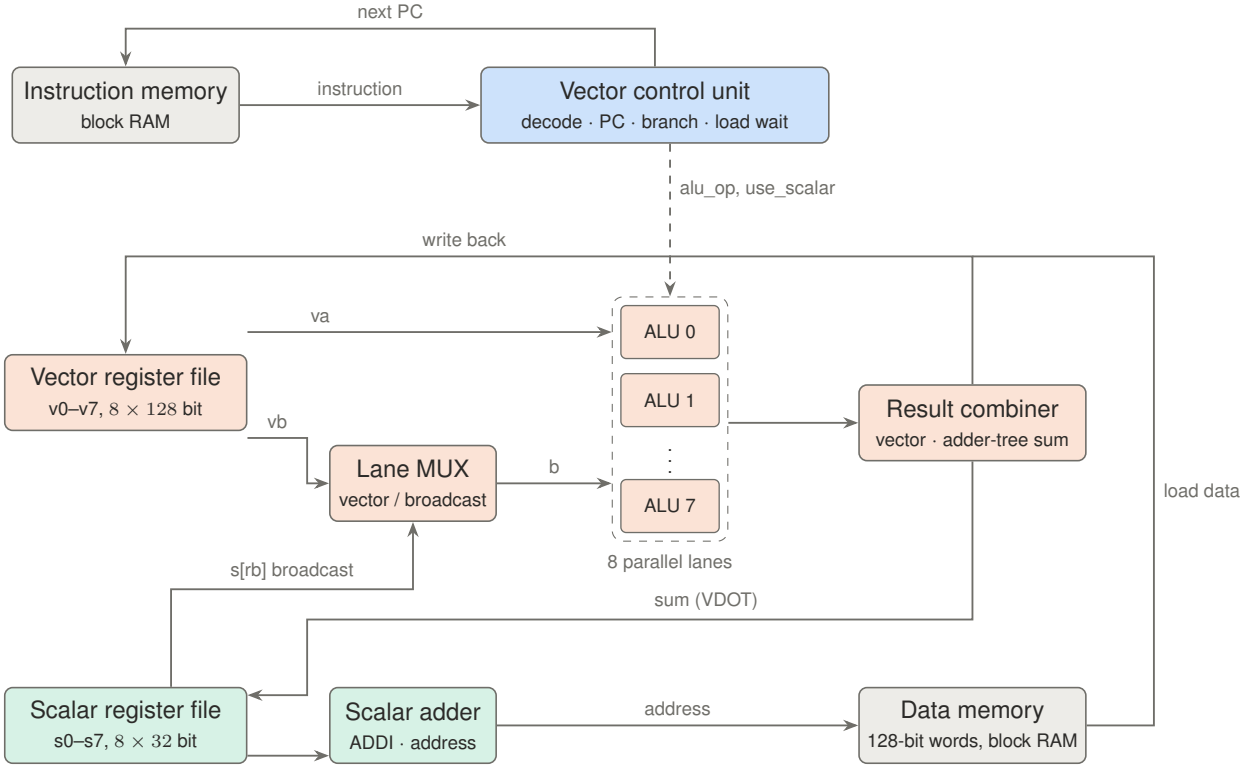
next PC
Instruction memory
block RAM
instruction
Vector control unit
decode · PC · branch · load wait
alu_op, use_scalar
write back
va
Vector register file
v0–v7, 8 × 128 bit
vb
Lane MUX
vector / broadcast
b
ALU 0
ALU 1
ALU 7
8 parallel lanes
Result combiner
vector · adder-tree sum
load data
s[rb] broadcast
sum (VDOT)
Scalar register file
s0–s7, 8 × 32 bit
Scalar adder
ADDI · address
address
Data memory
128-bit words, block RAM